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3031ICT: Applied Data Mining
3804ICT: Data Mining

Lecture Week 2: Data Exploration and Data Pre-processing



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Contents

- Data Exploration
  - Data Objects and Attribute Types
  - Basic Statistical Descriptions of Data
  - Data Visualization
  - Measurements of Data Similarity and Dissimilarity
- Data Pre-processing
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  - Data Integration
  - Data Transformation and Data Discretization
- Summary

Data Objects

- Data sets are made up of data objects.
- A **data object** represents an entity, also called *samples , examples, instances, data points, objects, tuples, observations.*
- Data objects are described by **attributes**.
- Database: rows -> data objects; columns ->attributes.

attributes.

objects

	age	workclass	fnlwgt	education	education-num	marital-status	occupation	relationship	race	sex	capital-gain	capital-loss	hours-per-week	native-country	income
0	39	State-gov	77516	Bachelors	13	Never-married	Adm-clerical	Not-in-family	White	Male	2174	0	40	United-States	<=50K
1	50	Self-emp-not-inc	83311	Bachelors	13	Married-civ-spouse	Exec-managerial	Husband	White	Male	0	0	13	United-States	<=50K
2	38	Private	215646	HS-grad	9	Divorced	Handlers-cleaners	Not-in-family	White	Male	0	0	40	United-States	<=50K
3	53	Private	234721	11th	7	Married-civ-spouse	Handlers-cleaners	Husband	Black	Male	0	0	40	United-States	<=50K
4	28	Private	338409	Bachelors	13	Married-civ-spouse	Prof-specialty	Wife	Black	Female	0	0	40	Cuba	<=50K

Attribute Types

- **Nominal Attributes**
  - are "symbols" or "names of things".
  - Each value represents some kind of category, code, or state, and so nominal attributes are also referred to as **categorical**. The values do not have any meaningful order.
  - **Examples:** ID numbers, eye color, zip codes
- **Ordinal Attributes**
  - have a meaningful order or *ranking* among them,
  - the magnitude between successive values is not known
  - **Examples:** rankings (e.g., taste of potato chips on a scale from 1-10), grades, height {tall, medium, short}
- **Interval-scaled attributes**
  - are measured on a scale of equal-size units.
  - have order and can be positive, 0, or negative.
  - providing a ranking of values and enable to quantify the *difference* between values.
  - **Examples:** calendar dates, temperatures in Celsius or Fahrenheit**.**
- **Ratio-scaled attributes**
  - are numeric with an inherent zero-point. That is, if a measurement is ratio-scaled, we can speak of a value as being a multiple (or ratio) of another value. In addition, the values are ordered, and we can also compute the difference between values, as well as the mean, median, and mode
  - **Examples:** temperature in Kelvin, length, time, counts

Discrete vs. Continuous Attributes

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- **Discrete Attribute**
  - Has only a finite or countably infinite set of values
    - E.g., zip codes, profession, or the set of words in a collection of documents
  - Sometimes, represented as integer variables
  - Note: Binary attributes are a special case of discrete attributes
- **Continuous Attribute**
  - Has real numbers as attribute values
    - E.g., temperature, height, or weight
  - Practically, real values can only be measured and represented using a finite number of digits
  - Continuous attributes are typically represented as floating-point variables

Data Exploration & Pre-processing

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Basic Statistical Descriptions of Data

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- Motivation
  - To better understand the data: central tendency, variation and spread
  - To pre-process your data well, it is essential to have an overall picture of data.
- Basic statistical descriptions
  - Measures of central tendency (where does the values of attribute most fall in?)
    - The location of the middle or centre of a data distribution
    - We will discuss mean, median, and midrange.
  - Measures of dispersion of the data (how does the values of attribute spread out?)
    - We will discuss range, quartiles, and interquartile range.
    - Five-number summary (min, 25%, median, 75%, max) and boxplot can be used.

Measures of Location: Mean and Median

- Mean:
  - The mean is the most common measure of the location of a set of points.
  - However, the mean is very sensitive to outliers.
  - Thus, the median or a trimmed mean is also commonly used.

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i \qquad \bar{x} = \frac{\sum_{i=1}^{n} w_i x_i}{\sum_{i=1}^{n} w_i}$$

- Median:
  - Middle value if odd number of values, or average of the middle two values otherwise

$$\text{median}(x) = \begin{cases} x_{(r+1)} & \text{if } m \text{ is odd, i.e., } m = 2r+1 \\ \frac{1}{2}(x_{(r)} + x_{(r+1)}) & \text{if } m \text{ is even, i.e., } m = 2r \end{cases}$$

- Midrange:
  - the average of the largest and smallest values in the set.

Measures of Spread: Range and Variance I

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- Range is the difference between the max and min (i.e. max-min)
- The variance or standard deviation s_x is the most common measure of the spread of a set of points.

$$\text{variance}(x) = s_x^2 = \frac{1}{m-1}\sum_{i=1}^{m}(x_i - \overline{x})^2$$

Sample Standard Deviation

- Because of outliers, other measures are often used.

$$\text{AAD}(x) = \frac{1}{m}\sum_{i=1}^{m}|x_i - \overline{x}|$$

$$\text{MAD}(x) = median\Big(\{|x_1 - \overline{x}|, \ldots, |x_m - \overline{x}|\}\Big)$$

$$\text{interquartile range}(x) = x_{75\%} - x_{25\%}$$

Measures of Spread: Range and Variance II

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- Range
  - difference between the largest (max()) and smallest (min()) values: max-min
- Quartiles
  - Quartiles are points taken at regular intervals of a data distribution, dividing it into essentially equal-sized consecutive sets. (Note: there may not be data values of X that divide the data into exactly equal-sized subsets, then round up to the next integer)
    - Q1 (25th percentile): splits off the lowest 25% of data from the highest 75%
    - Q3 (75th percentile): splits off the highest 25% of data from the lowest 75%
  - Inter-Quartile Range: IQR = Q3 – Q1
- Five number summary:
  - (min, Q1, median, Q3, max)
- Outlier: usually, a value more than 1.5 x IQR above Q3 (>Q3+1.5 x IQR) or below Q1 (<Q1-1.5 x IQR)
- Boxplot: ends of the box are the quartiles; median is marked; add whiskers, and plot outliers individually

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Visualization

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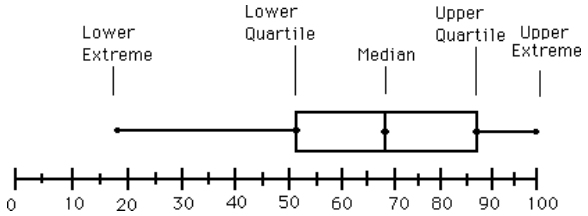
Visualization is the conversion of data into a visual or tabular format so that **the characteristics** of the data and **the relationship** among data items or attributes can be analyzed or reported.

- Visualization of data is one of the most powerful and appealing techniques for data exploration.
  - Humans have a well-developed ability to analyze a large amount of information that is presented visually
  - Can detect general patterns and trends
  - Can detect outliers and unusual patterns

Visualizations of Basic Statistical Descriptions

Boxplot: graphic display of five-number summary (another way of displaying the distribution of data)

- Five-number summary of a distribution (Minimum, Q1, Median, Q3, Maximum)

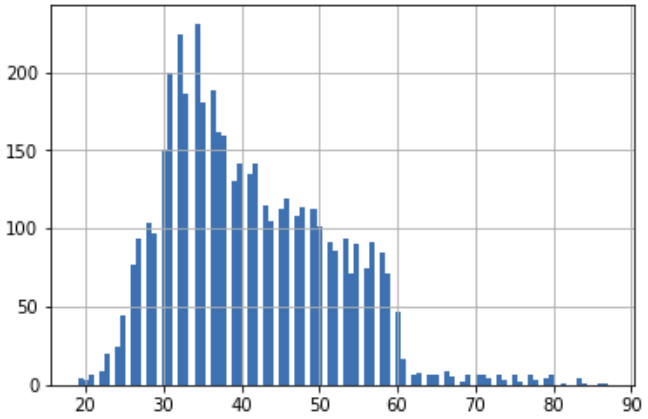


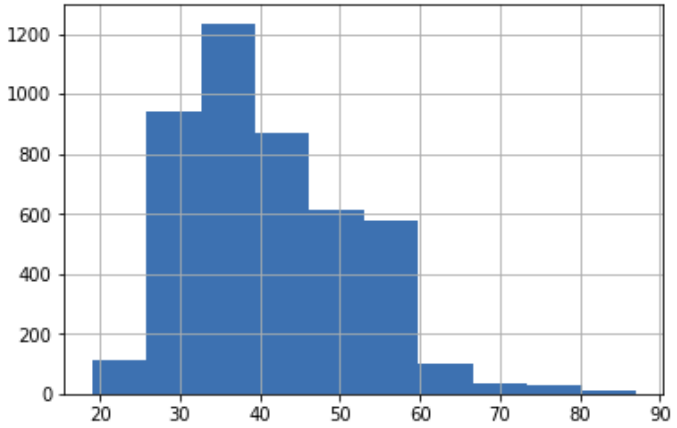
- Data is represented with a box
- The ends of the box are at the first and third quartiles, i.e., the height of the box is IQR
- The median is marked by a line within the box
- Whiskers: two lines outside the box extended to Minimum and Maximum
- Outliers: points beyond a specified outlier threshold, plotted individually

Visualization Techniques: Histograms

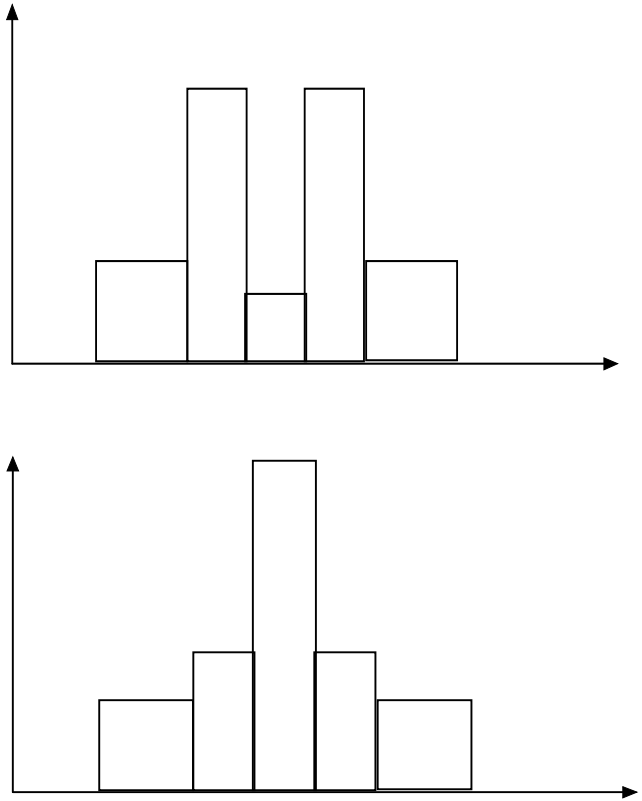
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- Histogram
  - Usually shows the distribution of values of a single variable
  - Divide the values into bins and show a bar plot of the number of objects in each bin.
  - The height of each bar indicates the number of objects
  - Shape of histogram depends on the number of bins
- Example: Age Histogram with different number of bins





Histograms Often Tell More than Boxplots



- The two histograms shown in the left may have the same boxplot representation
  - The same values for: min, Q1, median, Q3, max (five-number summary)
- But they have rather different data distributions, from histogram.

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Similarity and Dissimilarity

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- **Similarity**
  - Numerical measure of how alike two data objects are
  - Value is higher when objects are more alike
  - Often falls in the range [0,1]
- **Dissimilarity** (e.g., distance)
  - Numerical measure of how different two data objects are
  - Lower when objects are more alike
  - Minimum dissimilarity is often 0
  - Upper limit varies
- **Proximity** refers to a similarity or dissimilarity

Data Matrix and Dissimilarity Matrix

- Data matrix
  - n data points with p dimensions
  - n * p

$$\begin{bmatrix} x_{11} & \cdots & x_{1f} & \cdots & x_{1p} \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ x_{i1} & \cdots & x_{if} & \cdots & x_{ip} \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ x_{n1} & \cdots & x_{nf} & \cdots & x_{np} \end{bmatrix}$$

- Dissimilarity matrix
  - n data points, but registers only the distance
  - A symmetrical matrix
  - n * n

$$\begin{bmatrix} 0 & & & & \\ d(2,1) & 0 & & & \\ d(3,1) & d(3,2) & 0 & & \\ \vdots & \vdots & \vdots & & \\ d(n,1) & d(n,2) & \cdots & \cdots & 0 \end{bmatrix}$$

Dissimilarity between Nominal Variables

- Simple Matching:
  - *m*: # of matches, *p*: total # of variables

$$d(i,j) = \frac{p - m}{p}$$

- Example:

Name	Gender	Fever	Cough	Test-1	Test-2	Test-3	Test-4
Jack	M	Y	N	P	N	N	N
Mary	F	Y	N	P	N	P	N
Jim	M	Y	P	N	N	N	N

$$d(Jack, Mary) = \frac{2}{7} \qquad d(Jack, Jim) = \frac{2}{7} \qquad d(Mary, Jim) = \frac{4}{7}$$

Distance on Numeric Data: Minkowski Distance

- *Minkowski distance*: A popular distance measure

$$d(i, j) = \sqrt[h]{|x_{i1} - x_{j1}|^h + |x_{i2} - x_{j2}|^h + \cdots + |x_{ip} - x_{jp}|^h}$$

where $i = (x_{i1}, x_{i2}, \ldots, x_{ip})$ and $j = (x_{j1}, x_{j2}, \ldots, x_{jp})$ are two *p*-dimensional data objects, and *h* is the order (the distance defined is also called L-*h* norm)

- Properties
  - d(i, j) > 0 if i ≠ j, and d(i, i) = 0 (Positive definiteness)
  - d(i, j) = d(j, i) (Symmetry)
  - d(i, j) ≤ d(i, k) + d(k, j) (Triangle Inequality)
- A distance that satisfies these properties is a metric

Special Cases of Minkowski Distance

- *h* = 1: Manhattan (city block, L_1 norm) distance
  - E.g., the Hamming distance considers the number of bits that are different between two binary vectors

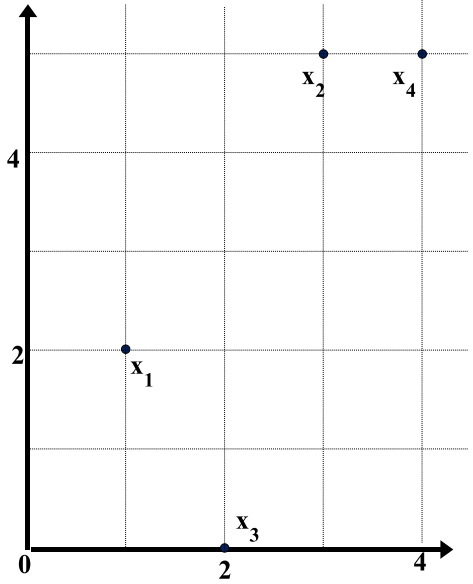
$$d(i,j) = |x_{i1} - x_{j1}| + |x_{i2} - x_{j2}| + \ldots + |x_{ip} - x_{jp}|$$

- *h* = 2: (L_2 norm) Euclidean distance

$$d(i,j) = \sqrt{(|x_{i1} - x_{j1}|^2 + |x_{i2} - x_{j2}|^2 + \ldots + |x_{ip} - x_{jp}|^2)}$$

Example: Minkowski Distance

point	attribute 1	attribute 2
x1	1	2
x2	3	5
x3	2	0
x4	4	5



Dissimilarity Matrices

Manhattan (L_1)

L	x1	x2	x3	x4
x1	0			
x2	5	0		
x3	3	6	0	
x4	6	1	7	0

$$d(i,j) = |x_{i1} - x_{j1}| + |x_{i2} - x_{j2}| + \ldots + |x_{ip} - x_{jp}|$$

Euclidean (L_2)

L2	x1	x2	x3	x4
x1	0			
x2	3.61	0		
x3	2.24	5.1	0	
x4	4.24	1	5.39	0

$$d(i,j) = \sqrt{(|x_{i1} - x_{j1}|^2 + |x_{i2} - x_{j2}|^2 + \ldots + |x_{ip} - x_{jp}|^2)}$$

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Data Quality

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- Poor data quality negatively affects many data processing efforts
- Data mining example: If a classification model for detecting people who are loan risky is built by using poor data.
  - Some credit-worthy candidates are denied loans
  - More loans are given to individuals that are not eligible
- Data quality problem:
  - Noise and outliers
  - Missing values
  - Duplicate data
  - Wrong data

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Data Cleaning

- Data in the Real World Is Dirty: Lots of potentially incorrect data, e.g., instrument faulty, human or computer error, transmission error
  - Incomplete: lacking attribute values, lacking certain attributes of interest, or containing only aggregate data
    - e.g., *Occupation*=“ ” (missing data)
  - Noisy: containing noise, errors, or outliers
    - e.g., *Salary*=“−10” (an error), *Age*=“200” (an error)
  - Inconsistent: containing discrepancies in codes or names, e.g.,
    - *Age*=“42”, *Birthday*=“03/07/2010”
    - Was rating “1, 2, 3”, now rating “A, B, C”
    - discrepancy between duplicate records
  - Intentional (e.g., *disguised missing* data)
    - Jan. 1 as everyone’s birthday?

Incomplete (Missing) Data

- Data is not always available
  - E.g., many tuples have no recorded value for several attributes, such as customer income in sales data
- Missing data may be due to
  - equipment malfunction
  - inconsistent with other recorded data and so deleted
  - data not entered due to misunderstanding
  - certain data may not be considered important at the time of entry
  - no register history or changes of the data
- Missing data may need to be inferred

How to Handle Missing Data?

- Ignore the tuple: usually done when class label is missing (when doing classification)—not effective when the % of missing values per attribute varies considerably
- Fill in the missing value manually: tedious + infeasible?
- Fill in it automatically with
  - a global constant : e.g., “unknown”, a new class?!
  - the attribute mean (e.g. “income” is missing)
  - the attribute mean for all samples belonging to the same class: smarter (e.g. “young”, “middle-aged”, “old”)
  - the most probable value: inference-based such as Bayesian formula or decision tree

Noisy Data

- Noise: random error or variance in a measured variable
- Incorrect attribute values may be due to
  - faulty data collection instruments
  - data entry problems
  - data transmission problems
  - technology limitation
  - inconsistency in naming convention
- Other data problems which require data cleaning
  - duplicate records
  - incomplete data
  - inconsistent data

How to Handle Noisy Data?

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- Binning
  - first sort data and partition into (equal-frequency) bins
  - then one can smooth by bin means, smooth by bin median, smooth by bin boundaries, etc.
- Regression (vs Classification, target label: Real Number vs Integer & Category)
  - smooth by fitting the data into regression functions
- Clustering (group data into clusters)
  - detect and remove outliers
- Combined computer and human inspection
  - detect suspicious values and check by human (e.g., deal with possible outliers, domain/expert knowledge)

Smoothing by Bins

- **Binning:** Binning methods smooth a sorted data value by consulting its "neighborhood," that is, the values around it. The sorted values are distributed into a number of "buckets" or *bins*. Because binning methods consult the neighborhood of values, they perform *local* smoothing.

Sorted data for *price* (in dollars): 4, 8, 15, 21, 21, 24, 25, 28, 34

Partition into (equal-frequency) bins:

Bin 1: 4, 8, 15
Bin 2: 21, 21, 24
Bin 3: 25, 28, 34

Smoothing by bin means:

Bin 1: 9, 9, 9
Bin 2: 22, 22, 22
Bin 3: 29, 29, 29

This figure illustrates some binning techniques. In this example, the data for *price* are first sorted and then partitioned into *equal-frequency* bins of size *(depth)* 3 (i.e., each bin contains three values).

Binning Methods for Data Smoothing

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- Sorted data for price (in dollars): 4, 8, 9, 15, 21, 21, 24, 25, 26, 28, 29, 34

* Partition into equal-frequency (**equal-depth**) bins:
  - Bin 1: 4, 8, 9, 15
  - Bin 2: 21, 21, 24, 25
  - Bin 3: 26, 28, 29, 34
* Smoothing by **bin means**:
  - Bin 1: 9, 9, 9, 9
  - Bin 2: 23, 23, 23, 23
  - Bin 3: 29, 29, 29, 29

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Data Integration

- Data integration (Aggregation):
  - Combines data from multiple sources into a coherent store
- Schema integration: e.g., A.cust-id ≡ B.cust-#
  - Integrate metadata from different sources
- Entity identification problem:
  - Identify real world entities from multiple data sources, e.g., Can Wang = Volcano Wang
- Detecting and resolving data value conflicts
  - For the same real-world entity, attribute values from different sources are different
  - Possible reasons: different representations, different scales (e.g., Metric vs. British Imperial units)

Handling Redundancy in Data Integration

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- Redundant data often occur when integrating multiple databases
  - *Object identification*: The same attribute or object may have different names in different databases (e.g. “nappy” and “diaper”)
  - *Derivable data:* One attribute may be a “derived” attribute in another table (e.g., annual revenue from quarterly revenue)

- Redundant attributes may be able to be detected by *correlation analysis* and *covariance analysis*

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Data Transformation

- A function that maps the entire set of values of a given attribute to a new set of replacement values, so that the resulting mining process may be more efficient or effective.
- Methods
  - Smoothing
  - Attribute/feature construction
  - Aggregation
  - Normalization
    - min-max normalization
    - z-score normalizations
  - Discretization

Normalize Numerical Attributes

- **Min-max normalization**: to [new_min$_A$, new_max$_A$]

$$v' = \frac{v - min_A}{max_A - min_A}(new_max_A - new_min_A) + new_min_A$$

  - Ex. Let income range \$12,000 to \$98,000 normalized to [0.0, 1.0]. Then \$73,600 is mapped to $\frac{73,600 - 12,000}{98,000 - 12,000}(1.0 - 0) + 0 = 0.716$

- **Z-score normalization** (μ: mean, σ: standard deviation):

$$v' = \frac{v - \mu_A}{\sigma_A}$$

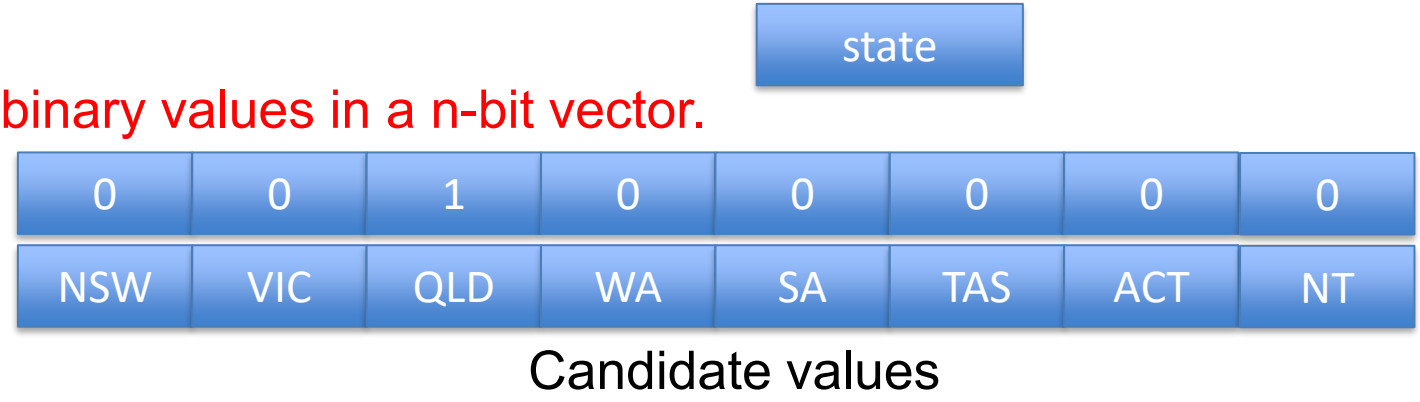
  - Ex. Let μ = 54,000, σ = 16,000. Then

$$\frac{73,600 - 54,000}{16,000} = 1.225$$

Normalize Categorical Attributes

- It is often to see some categorical features, nominal or ordinal:
  - Example: Gender (F and M), country (AUS, USA, CHN, etc.), state (NSW, VIC, QLD, etc.)
- How can we use them if we convert them into integer:
  - Example: Gender: 0 or 1, country: 0, 1, 2, …, state: 0, 1, 2, 3, …
- One-Hot encoding can be used to transform categorical features:

binary values in a n-bit vector.



E.g: original value of state is 2 (or QLD), then the one-hot encoding is [0,0,1,0,0,0,0,0]

Discretization

- Discretization: Divide the range of a continuous attribute into intervals
  - E.g. to replace the age by interval labels 0–10, 11–20, etc., with conceptual labels youth, adult, senior.
- Interval labels can then be used to replace actual data values, the size of data is thus reduced.
- Typical methods: All the methods can be applied recursively
  - Binning
    - Top-down split, unsupervised
  - Histogram analysis
    - Top-down split, unsupervised
  - Clustering analysis (unsupervised, top-down split or bottom-up merge)
  - Decision-tree analysis (supervised, top-down split)
  - Correlation (e.g., χ^2) analysis (unsupervised, bottom-up merge)

Simple Discretization: Binning

- Equal-width (distance) partitioning
  - Divides the range into *N* intervals of equal size: form uniform grid
  - if *A* and *B* are the lowest and highest values of the attribute, the width of intervals will be: $W = (B - A)/N$, *N* is the number of bins.
  - The most straightforward, but outliers may dominate presentation
  - Skewed data is not handled well
- Equal-depth (frequency) partitioning
  - Divides the range into *N* intervals, each containing approximately same number of samples
  - Good data scaling
  - Managing categorical attributes can be tricky

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  - Data Cleaning
  - Data Integration
  - Data Reduction
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Have a question?

